UNIT 5: 3D Objects Representation(7Hrs)

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Bsc.CSIT(3rd)

Introduction

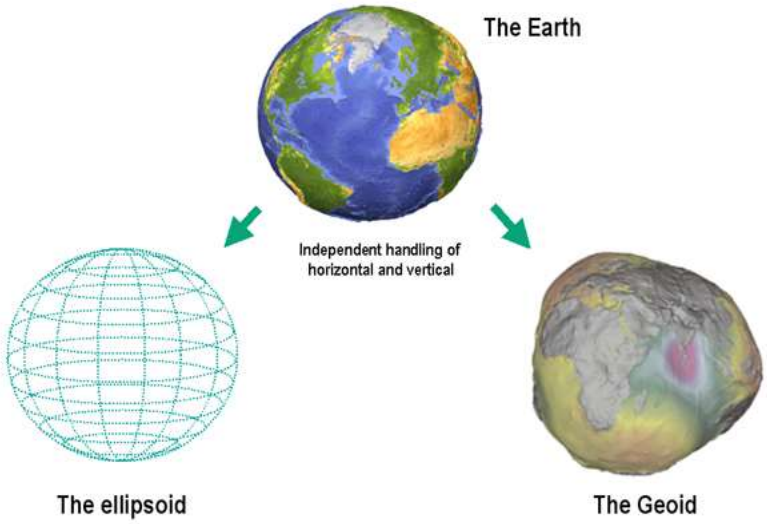
- Graphics scenes ***can contain many different kinds of objects***: trees , flowers, clouds, rocks, water, bricks, wood paneling, rubber, paper, marble, steel, glass, plastic, and cloth, just to mention a few.
- ***No single method can use to describe objects*** that will include all characteristics of these different materials.
- And to produce realistic displays of scenes, we need to use representations that accurately model object characteristics.
  - ***Polygon and quadric surfaces*** provide precise descriptions for simple Euclidean objects ***such as polyhedrons and ellipsoids***;
  - ***Spline surfaces*** are useful for ***designing aircraft wings, gears, and other engineering*** structures with curved surfaces;

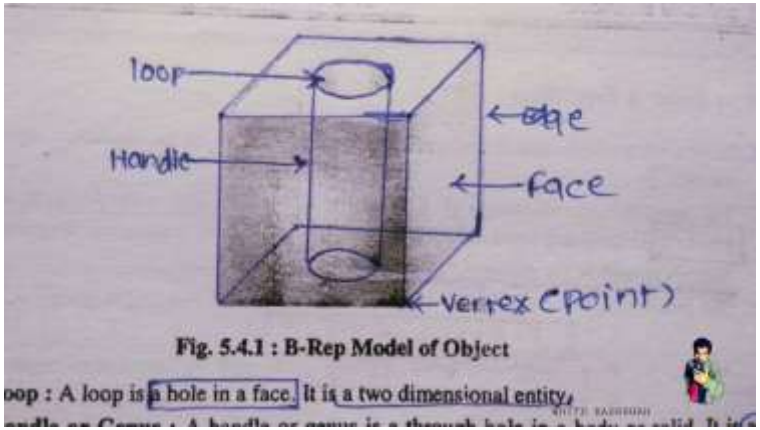
Introduction

- Representation schemes for solid objects are often divided into two broad categories:
  - ***Boundary representations (B-reps)***
  - ***Space-partitioning representations***
- B-reps describe a three-dimensional object as a ***set of surfaces*** that ***separate the object interior from the environment***. Typical examples of boundary representations are polygon facets and spline patches.
- Space-partitioning representations are used to describe interior properties, ***by partitioning the spatial region containing an object into a set of small, nonoverlapping, contiguous solids*** (usually **cubes).**

Boundary Representation

- Boundary representation (B-rep) is a fundamental technique in computer graphics for modeling 3D objects. It defines the shape of an object by describing its outer surfaces, including the vertices, edges, and faces that make up its boundary.
- The solids are described by its boundary surface. Uses the description by vertices, edges and faces. - The most common representation is the boundary polygons.
- E.g. Polyhedron, ellipsoid, aircraft, medical images etc.

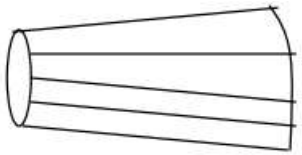




Boundary Representation

- Each 3D object is supposed to be formed its surface by collection of polygon facets and spline patches. Some of the boundary representation methods for 3D surface are:

1. **Polygon Surfaces:** It is the most common representation for 3D graphics object. In this representations, a 3D object is represented by a set of surfaces that enclose the object interior. Many graphics system use this method. Set of polygons are stored for object description. This simplifies and speeds up the surface rendering and display of object since all surfaces can be described with linear equations.



A 3D object represented by polygons

The polygon surface are common in design and solid-modeling applications, since wire frame display can be done quickly to give general indication of surface structure. Then realistic scenes are produced by interpolating shading patterns across polygon surface to illuminate.
**There are three way to represent polygon surface: Polygon Table, Plane Equations, & Polygon Meshes.

Polygon Surfaces

- **Polygon Table:** A polygon surface is specified with a set of vertex co-ordinates and associated attribute parameters. A convenient organization for storing geometric data is to create 3 lists:
- A vertex table
- An edge table
- A polygon surface table.

− Vertex table stores co-ordinates of each vertex in the object.

− The edge table stores the Edge information of each edge of polygon facets.

− The polygon surface table stores the surface information for each surface i.e. each surface is represented by edge lists of polygon.

-Are used to create video games, films, and virtual reality environments

**attribute tables store information about objects, while geometric tables store information about the shape of objects.

Polygon Surfaces

Attribute tables:

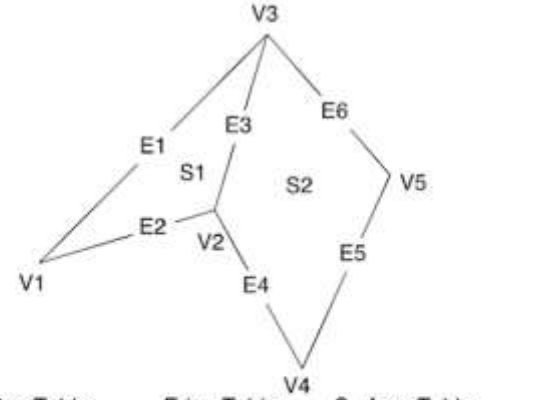
- ontain information about an object, such as its transparency, reflectivity, and texture
- Are organized into rows and columns, with each row representing a feature and each column representing an attribute
- Can be used to store information about geographic features, such as cities, with attributes like name, population, and postal code.
- Attribute tables are used to store information about objects in GIS

Polygon Surfaces

- Guidelines ***to generate Error Free tables:***
  1) Every vertex is listed as an endpoint for at least two edges,
  2) Every edge is part of at least one polygon,
  3) Every polygon is closed,
  4) Each polygon has at least one shared edge, and
  5) If the edge table contains pointers to polygons, every edge referenced by a polygon pointer has a reciprocal pointer back to the polygon.

Polygon Surfaces

- Consider the surface contains polygonal facets as shown in figure (only two polygon are taken here)
- S1 and S2 are two polygon surface that represent the boundary of some 3D object. Foe storing geometric data, we can use following three table.
- The object can be displayed efficiently by using data from tables and processing them for surface rendering and visible surface determinatior



Vertex Table

V1 : x1,y1,z1
V2 : x2,y2,z2
V3 : x3,y3,z3
V4 : x4,y4,z4
V5 : x5,y5,z5

Edge Table

E1 : V1, V3
E2 : V1, V2
E3 : V2, V3
E4 : V2, V4
E5 : V4, V5
E6 : V5, V3

Surface Table

S1 : V1, V2, V3
S2 : V2, V4, V5, V3

or

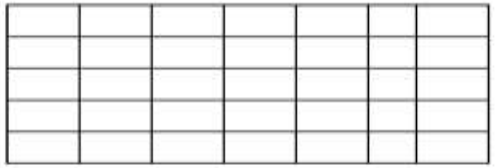
S1 : E1, E2, E3
S2 : E3, E4, E5, E6

Polygon Surfaces

- **Polygon Meshes:** A polygon mesh is collection of edges, vertices and polygons connected such that each edge is shared by at most two polygons. An edge connects two vertices and a polygon is a closed sequence of edges. An edge can be shared by two polygons and a vertex is shared by at least two edges. When object surface is to be tiled, it is more convenient to specify the surface facets with a mesh function. One type of polygon mesh is triangle strip. This function produce n-2 connected triangles.

Triangular Mesh

Another similar function is the quadrilateral mesh, which generates a mesh of (n-1) by (m-1) quadrilaterals, given the co-ordinates for an n by m array of vertices



6 by 8 vertices array , 35 element quadrilateral mesh

- If the surface of 3D object is planer, it is comfortable to represent surface with meshes.

Polygon Meshes

Explicit Representation:

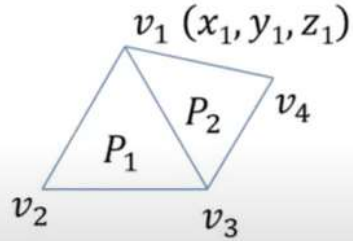
In explicit representation each polygon stores all the vertices in order in the memory as,

$$P = (((x1, y1, z1), (x2, y2, z2)), \ldots, ((xm, ym, zm), (xn, yn, zn)))$$

The mesh is represented by a list of vertex coordinates.

It process fast but requires more memory for storing.

- Example:



$$P_1 = \begin{pmatrix} ((x_1, y_1, z_1), (x_2, y_2, z_2)), ((x_2, y_2, z_2), (x_3, y_3, z_3)), \\ ((x_3, y_3, z_3), (x_1, y_1, z_1)) \end{pmatrix}$$

Pointer to Vertex List-Polygon Meshes

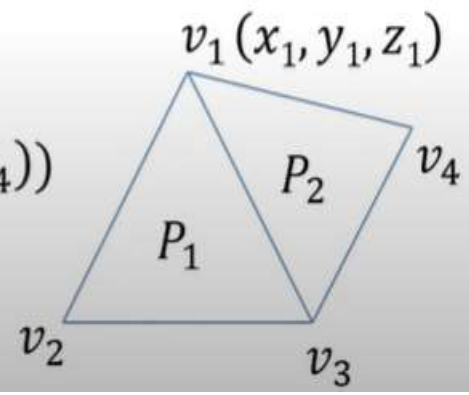
- In this method, each vertex stores in vertex list and then polygon contains pointer to the required vertex.

$$V = ((x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_n, y_n, z_n))$$

- Now polygon of vertices 3, 4, 5 is represented as:

$$P = ((v_3, v_4), (v_4, v_5), (v_5, v_3))$$

- It is considerably space saving but for finding common edges we require extra work.
- Example:
- $V = ((x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_4, y_4, z_4))$
- $P_1 = ((v_1, v_2), (v_2, v_3), (v_3, v_1))$
- $P_2 = ((v_1, v_3), (v_3, v_4), (v_4, v_1))$

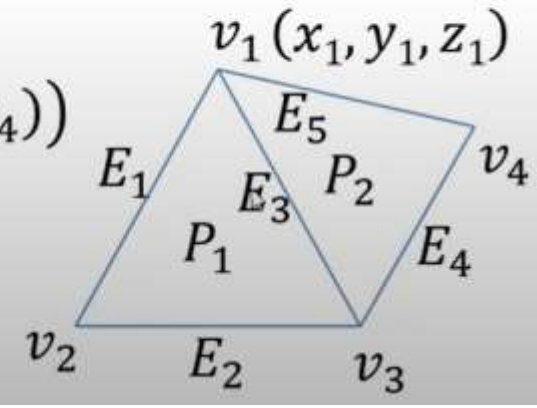


Pointer to Edge List-Polygon Meshes

- In this method, polygon have pointers to the edge list and edge list have pointer to vertex list for each edge two vertex pointer is required which points in vertex list.

$$V = ((x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_n, y_n, z_n))$$
$$E = ((v_1, v_2), (v_2, v_3), \dots, (v_m, v_n))$$
$$P = (E_1, E_2, \dots, E_k)$$

- This approach is more memory efficient and easy to find common edges. For example,
- $V = \big((x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_4, y_4, z_4)\big)$
- $E = ((v_1, v_2), (v_2, v_3), \dots, (v_4, v_1))$
- $P_1 = (E_1, E_2, E_3)$



Plane Equation Method

Plane Equation Method:

Plane equation method is another method for representing the polygon surface for 3D - object. The information about the spatial orientation of object is described by its individual surface which is obtained by the vertex co-ordinate values and the equation of each plane gives the description of each surface. The equation for a plane surface can be expressed in the form

$$Ax + By + Cz + D = 0$$

Where (x, y, z) is any point on the plane, and A, B, C, D are constants describing the spatial properties of the plane. The values of A, B, C, D can be obtained by solving a set of three plane equations using co-ordinate values of 3 non collinear points on the plane.

Let $(x_1, y_1, z_1), (x_2, y_2, z_2)$ and (x_3, y_3, z_3) are three such points on the points on the plane. Then

$$Ax_1 + By_1 + Cz_1 + D = 0 \quad \text{———} \quad (1)$$

$$Ax_2 + By_2 + Cz_2 + D = 0 \quad \text{———} \quad (2)$$

$$Ax_3 + By_3 + Cz_3 + D = 0 \quad \text{———} \quad (3)$$

The solution of these equations can be obtained in determinant form using Cramer's rule as.

$$A = \begin{vmatrix} 1 & y_1 & z_1 \\ 1 & y_2 & z_2 \\ 1 & y_3 & z_3 \end{vmatrix} \qquad B = \begin{vmatrix} x_1 & 1 & z_1 \\ x_2 & 1 & z_2 \\ x_3 & 1 & z_3 \end{vmatrix}$$

$$C = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \qquad D = -\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

For any point (x, y, z)

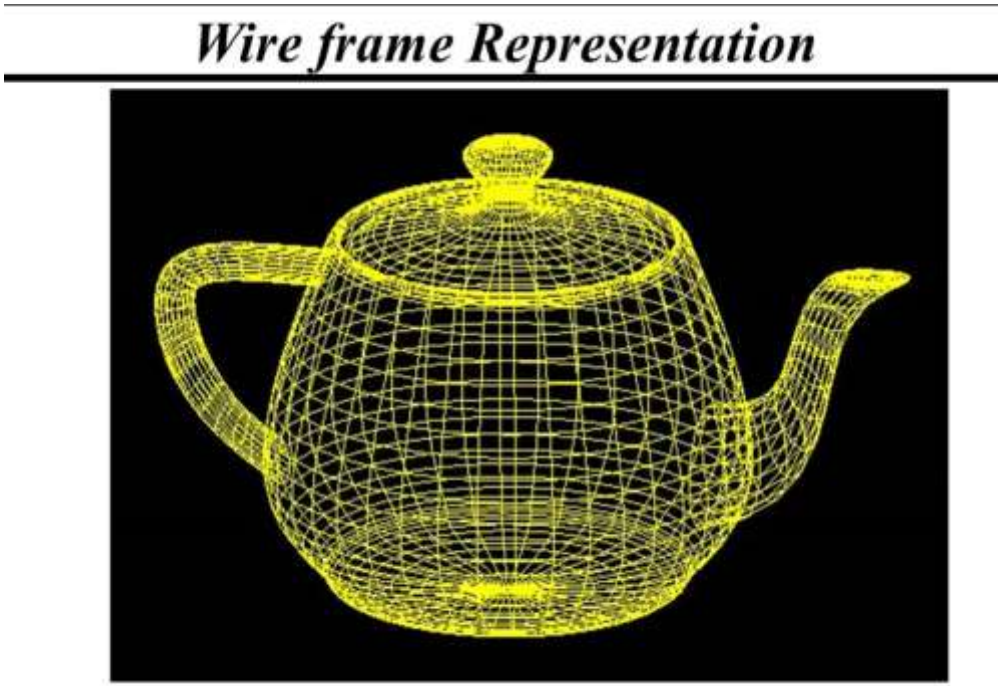
if $Ax + By + Cz + D \neq 0$ then (x, y, z) is not on the plane

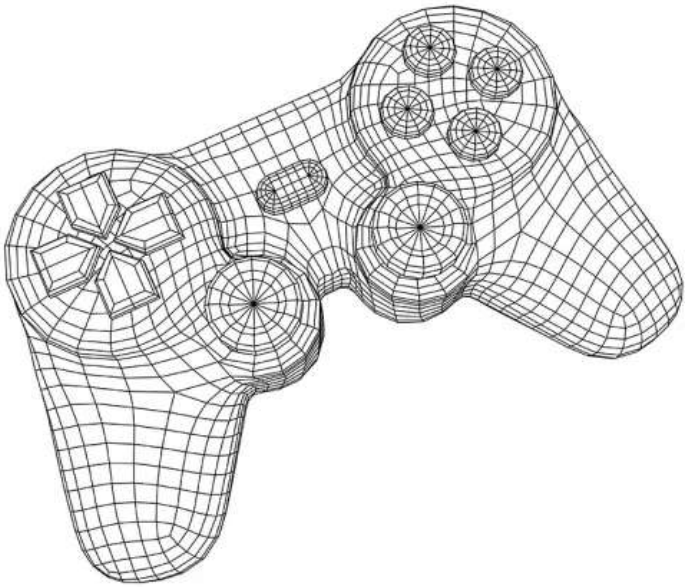
if $Ax + By + Cz + D < 0$, i.e. invisible side the point (x, y, z) is inside plane

if $Ax + By + Cz + D > 0$ the point lies outside the surface.

Wireframe Representation:

In this method 3D objects is represented as a list of straight lines, each of which is represented by its two end points (x1,y1,z1) and (x2,y2,z2). This method only shows the skeletal structure of the objects. It is simple and can see through the object and fast method. But independent line data structure is very inefficient i.e. don't know what is connected to what. In this method the scenes represented are not realistic.





Wireframe Representation:

- **Advantages of Wire Frame Model :-**

1. It is simple methods and requires less memory space.
2. It forms the basis for surface and solid modeling.
3. Manipulations in the model can be done easily and quickl.

Wireframe Representation:

Disadvantages :-

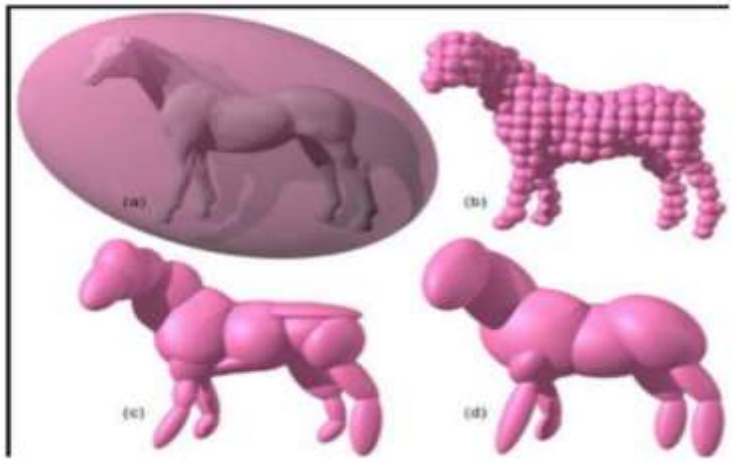
1. One of the serious limitation in the ambiguity of orientation and viewing plane
2. Cannot model complex curve surfaces
3. Does not represent an actual solid (no surface and volume)
4. Physical properties such as mass, surface area, volume, center of gravity etc. are not possible to calculate
5. Wireframe models has no knowledge of surface faces, therefore it will not detect interface between two matting components and this is serious drawback especially in component assembly, kinematic analysis, NC tool and robot arm simulation.

Polygon Surfaces

Blobby Objects: Some objects don't maintain a fixed shape but change their surface characteristics in certain motions or when proximity to another objects e.g molecular structures, water droplets, other liquid effects, melting objects, muscle shaped in human body etc. These objects can be described as exhibiting "blobbiness" and are referred as blobby objects. Several models have been developed for representing blobby objects as distribution functions over a region of space.

- One way is to use Gaussian density function or bumps.
- A surface function is defined as:

Blobby Objects



Polygon Surfaces

$$f(x,y,z) = \sum_k b_{k0} e^{-a_k r_k^2} - T = 0$$

Where $r_k = \sqrt{x_k^2 + y_k^2 + z_k^2}$, T = Threshold and a and b are used to adjust amount of blobbness.

Other method for generating for generating blobby objects uses quadratic density function as:

$$f(x) = \begin{cases} b(1 - \frac{3r^3}{d^2}) \\ \frac{3}{2} b(1 - \frac{r}{d})^2 \\ 0 \end{cases}$$

Advantages

- Can represent organic, blooby or liquid line structures
- Suitable for modeling natural phenomena like water, human body
- Surface properties cam be easily derived from mathematical equations.

Disadvantages:

- Requires expensive computation
- Requires special rendering engine
- Not supported by most graphics hardware

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Representing Curves:

Spline Representation

A Spline is a flexible strips used to produce smooth curve through a designated set of points.

A curve drawn with these set of points is spline curve. Spline curves are used to model 3D object surface shape smoothly.

Mathematically, spline are described as piece-wise cubic polynomial functions. In computer graphics, a spline surface can be described with two set of orthogonal spline curves. Spline is used in graphics application to design and digitalize drawings for storage in computer and to specify animation path. Typical CAD application for spline includes the design of automobile bodies, aircraft and spacecraft surface etc.

Representing Curves:

<u>Interpolation and approximation spline</u>

- Given the set of control points, the curve is said to interpolate the control point if it passes through each points.
- If the curve is fitted from the given control points such that it follows the path of control point without necessarily passing through the set of point, then it is said to approximate the set of control point.

Parametric cubic Curves

- Parametric splines are defined by a set of equations that represent a curve or surface. These equations can be used to generate points along the spline, which determines its shape. Parametric splines are commonly used in computer graphics, where they are used to model 3D objects and animations.
- There are many different types of parametric splines, including Bezier curves, B-splines etc. Parametric curves are mathematical representations of curves where the coordinates of points on the curve are defined by one or more parameters.

Parametric continuity condition

- For smooth transition from one curve section on to next curve section we put various continuity conditions at connection points.

Let parametric coordinate functions as

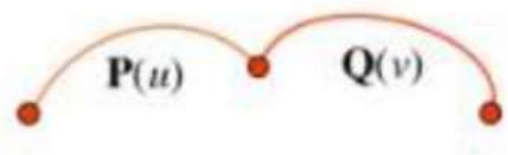
$$x = x(u),\ y = y(u),\ z = z(u) \qquad \because u1 \ll u \ll u2$$

Parametric cubic Curves

1. <u>Zero Order Parametric Continuity</u>

   (C^0)

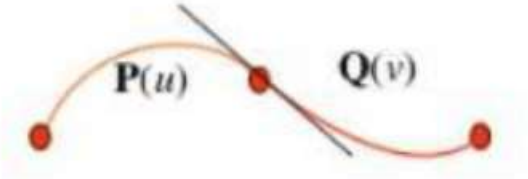
It meant simply that the curve meets i.e., the values of (x, y, z) evaluated at u_2 for the first curve section is equal to the (x, y, z) values of u_1 for the next curve section.



2. <u>First Order Parametric Continuity</u>

   (C^1)

It means the first parametric derivative (tangent lines) of the coordinate function for two successive crime sections are equal at their joining point.



Parametric cubic Curves

3. <u>Second Order Parametric Continuity</u>

   (C^2)

   It means that both first and second derivative of the coordinate function for two consecutive crime sections are same at their intersection.

Representing Curves:

- **Cubic Spline:**
- Cubic Splines are mostly used for representing path of moving object or existing object shape or drawing.
- It is used for design the object shapes.
- Cubic polynomial offer a reasonable compromise between flexibility and speed of computation. Cubic spline require less calculations compares to higher order polynomials and less memory.
- Given a set of control points, cubic interpolation splines are obtained by fitting the input points with a piecewise cubic polynomial curve that passes through every control points.
- Suppose we have n+1 control points specified with co-ordinates
- pk = (xk , yk ,zk),k = 0,1,2.....n A cubic interpolation fit of these points is

We can describe the parametric cubic polynomial that is to be filled between each pair of control points with the following set of equations.

$$x(u) = a_x u^3 + b_x u^2 + c_x u + dx$$
$$y(u) = a_y u^3 + b_y u^2 + c_y u + dy \qquad (0 \le u \le 1)$$
$$z_u(u) = a_z u^3 + b_z u^2 + c_z u + dz$$

Representing Curves:

- For above equation we need to determine for constant a, b and c and d the polynomial representation for each of n curve section.

This is obtained by setting proper boundary condition at the joints.

Common method for setting this conditions are,

1. Natural Cubic splines
2. Hermit interpolation
3. Cardinal Splines
4. Kochanek-Bartels spline

Hermit Interpolation

- Hermite curve named after the French mathematician Charles Hermite is an interpolating piecewise cubic polynomial. It has a specified tangent at each control point.
- The **Hermite curve** in computer graphics is an interpolation spline curve.
- Hermite spline curves can be adjusted locally because each section is only dependent on its endpoint constraints.

Properties of Hermite Curve

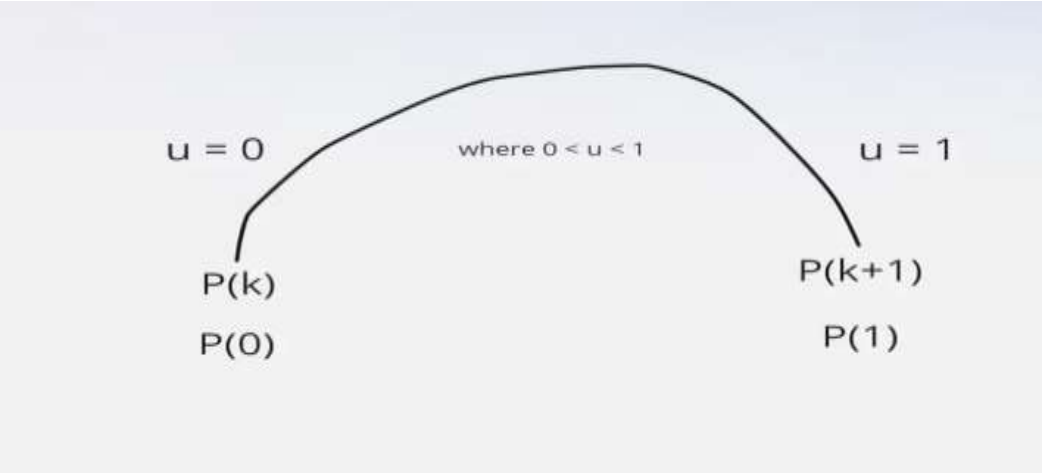
- **Interpolation:** Hermit curves interpolate smoothly between their control points.
- **Tangent Control:** Hermit curves allow precise control over the tangent vectors at each control point.
- **Parametric Representation:** Hermite curves are often expressed as parametric equations, where a parameter u varies between 0 and 1.
- **Derivatives:** Hermite curves have continuous first and second derivatives.
- **Local Control:** Changes made to one segment of a Hermite curve affect only that segment, providing local control.
- **Polynomial Form:** The Hermite curve is expressed as a polynomial function, typically a cubic polynomial.
- **Versatility in Applications:** Hermite curves find applications in computer graphics, computer-aided design (CAD), animation, and modeling.

Hermit Interpolation

- Parametric cubic point function for any curve section is,

$$P(u) = au^3 + bu^2 + cu + d$$
$$\text{where } 0 < u < 1$$

The above equation is very important we will use this equation in further steps below. Let's Understand Hermite Curve Derivation,



Hermit Interpolation

- In the above figure, we let two variables i.e.
- **P (k) which is P(0), and P (k+1) which is P(1).**
- Now we have to let the **derivative of P(k) which is D Pk.**
- and **derivative of P(k+1) which is D Pk+1.**

$$p(0) = p_k$$
$$p(1) = p_{k+1}$$
$$p'(0) = dp_k$$
$$p'(1) = dp_{k+1}$$

Now we have to find the **derivative of the Hermite curve Mathematical Expression** as follows:

$$P(u) = au^3 + bu^2 + cu + d$$

$$P'(u) = 3au^2 + 2bu + c + d$$

$$\text{where } 0 < u < 1$$

After finding the derivative of P(u) **we need to put the values 0 and 1 in the u parameter in both the above equations** as follows:

Hermit Interpolation

$$P(u) = au^3 + bu^2 + cu + d$$

$$P(k) = P(0) \quad a.(0)^3 + b.(0)^2 + c.0 + d.0$$

$$= a.0 + b.0 + c.0 + d.0 \longrightarrow (1)$$

$$P(k+1) = P(1) \quad a.(1)^3 + b.(1)^2 + c.1 + d.1$$

$$= a.1 + b.1 + c.1 + d.1 \longrightarrow (2)$$

$$P'(u) = 3au^2 + 2bu + c + d$$

$$D\,Pk = P(0) \quad 3a.(0)^3 + 2b.(0) + c.1 + d.0$$

$$= a.0 + b.0 + c.1 + d.0 \longrightarrow (3)$$

$$DP(k+1) = P(1) \quad 3a.(1)^2 + 2b.(1) + c.1 + d.0$$

$$= a.3 + b.2 + c.1 + d.0 \longrightarrow (4)$$

Hermit Interpolation

- So after putting u as 0,1, we have 4 equations as follows:

$$= a.0 + b.0 + c.0 + d.0 \longrightarrow (1)$$

$$= a.1 + b.1 + c.1 + d.1 \longrightarrow (2)$$

$$= a.0 + b.0 + c.1 + d.0 \longrightarrow (3)$$

$$= a.3 + b.2 + c.1 + d.0 \longrightarrow (4)$$

Now we have to represent these equations in terms of matrix as shown below so that we can find the Hermite matrix.

Hermit Interpolation

$$\begin{bmatrix} P(0) \\ P(1) \\ P'(0) \\ P'(1) \end{bmatrix} or \begin{bmatrix} P_k \\ P_{k+1} \\ DP_k \\ DP_{K+1} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$$

Now we can calculate the value of a, b, c by taking the inverse of the equation matrix. The formula for calculating the inverse of the 4x4 matrix is **A inverse= adj(A)/det(A)**

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} P_k \\ P_{k+1} \\ DP_k \\ DP_{K+1} \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} P_k \\ P_{k+1} \\ DP_k \\ DP_{K+1} \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 2 & -2 & 1 & 1 \\ -3 & 3 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_k \\ P_{k+1} \\ DP_k \\ DP_{K+1} \end{bmatrix}$$

Hermit Interpolation

- So the above figure is a **Hermite matrix** after finding the Hermite matrix we need to multiply these two matrixes on the right side to find the value of a, b, c, d.

$$P(u) = Pk(2u3 - 3u2 + 1) + Pk + 1(-2u3 + 3u2) + DPk(u3 - 2u2 + u) + DPk + 1(u3 - u2)$$

$$P(u) = PkH0(u) + Pk + 1H1(u) + DPkH2(u) + DPk + 1H3(u)$$

Representing Curves:

Bezier Curve and surface:

This spline approximation method was developed by the French Engineer Pierre Bezier for use in the design of automobile body.

Bezier splines have a no of properties that make them highly useful and convenient for curve and surface design. They are easy to implement. For this reason, Bezier spline are widely available in various CAD systems.

In General Bezier curve can be fitted to any number of control points. The no of control points to be approximated and their relative position determine the degree of Bezier polynomial.

The Bezier curve can be specified with boundary conditions, with characterizing matrix or blending functions. But for general blending function specification is most convenient.

Bezier Curve and surface:

suppose we have n+1 control points positions $p_k = (x_k, y_k, z_k), k = 0,1,2,....n$. These co-ordinate points can be blended to produce the following position vector p(u) which describes path of and approximating Bezier polynomial function between p_o and p_n.

$$p(u) = \sum_{k=0}^{n} p_k BEZ_{k,n}(u) \quad 0 \le u \le 1 \quad \text{———} \quad (1)$$

The Bezier blending functions $BEZ_{k,n}(u)$ the Bernstein polynomial.

$BEZ_{k,n}(u) = C(n,k)u^k(1-u)^{n-k}$ where

$$C(n,k) = \frac{n!}{(n-k)!k!}$$

The vector equation (1) represents a set of three parametric equations for individual curve condition

$$x(u) = \sum_{k=0}^{n} x_k BEZ_{k,n}(u)$$

$$y(u) = \sum_{k=0}^{n} y_k BEZ_{k,n}(u)$$

$$z(u) = \sum_{k=0}^{n} z_k BEZ_{k,n}(u)$$

- Bezier curve is a polynomial of degree one less than control points.

Limitation Of Bezier Spline

- Bezier spline limitations include limited shape representation (sharp corners are difficult), global control (moving one point affects the entire curve).
- The no of control points result a cubic polynomial curve, so to reduce the degree of polynomial we need to reduce the no. control points & vice versa.
- **Every Control Point Affects the Entire Curve:** (A change to any control point can affect the entire curve, making it difficult to achieve local control (where changes to one area don't affect others).)
- **Limited Continuity:** (Bezier curves may not always achieve smooth continuity between adjacent curves, potentially resulting in visible discontinuities or sharp transitions.)
- **Requires Careful Arrangement of Control Points:** (To achieve desired levels of continuity, control points need to be carefully arranged, which can reduce local control.)

B-Spline

- B-spline curve is local control. {note: does not depend on control point}
- It depend on the order of polynomial.
- Each vertex affects the shape on the limited range of parameter.
- In B-spline curve the degree of the curve and the no of vertices are independent.
- Thus it is possible to change the degree of curve without changing the no of vertices of the polygon.

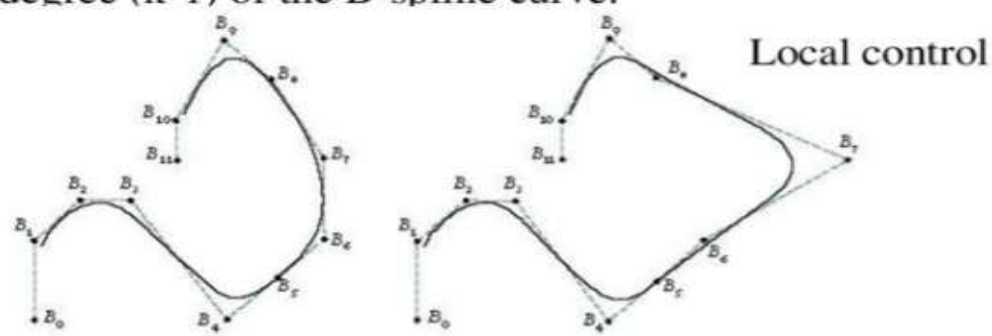
B-Spline

- B-spline blending functions, also known as basis functions, are mathematical functions that combine control points to create smooth curves or surfaces. They are defined recursively and are crucial for generating B-spline curves and surfaces.

The B-spline curve defined by n+1 control points P_i consists of $n-2$ curve segments and is given by:

$$\mathbf{P}(u) = \sum_{i=0}^{n} \mathbf{P}_i \cdot N_{i,k}(u), \qquad 0 \le u \le u_{max}$$

where $N_{i,k}(u)$ are the B-spline (blending or <u>basis</u>) functions. The parameter k controls the degree (k-1) of the B-spline curve.



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$$N_{i,k}(u) = (u - u_i)\frac{N_{i,k-1}(u)}{u_{i+k-1} - u_i} + (u_{i+k} - u)\frac{N_{i+1,k-1}(u)}{u_{i+k} - u_{i+1}}$$

where

$$N_{i,1} = \begin{cases} 1, & u_i \le u \le u_{i+1} \\ 0, & \text{otherwise} \end{cases}$$

B-spline Curves: Important Properties

- The degree of B-spline curve polynomial does not depend on the number of control points which makes it more reliable to use than Bezier curve.
- B-spline curve provides the local control through control points over each segment of the curve.
- We can add/modify any no of control points to change the shape of the curve without affecting the degree of polynomial.
- Approximation spline curve
- Blending function not necessary to be non zero.

Advantages of B-splines:

- **Local Control:** B-splines allow for local control, meaning that modifying a single control point only affects a limited portion of the curve or surface.
- **Flexibility:** B-splines can represent a wide variety of shapes and surfaces.
- **Mathematical Foundation:** B-splines have a solid mathematical foundation, making them a reliable tool for geometric modeling.

Representing Curves:

Quadric Surface

Quadric Surface is one of the frequently used 3D objects surface representation. The quadric surface can be represented by a second degree polynomial. This includes:

1. Sphere: For the set of surface points (x,y,z) the spherical surface is represented as:
   $x^2+y^2+z^2 = r^2$, with radius r and centered at co-ordinate origin.

2. Ellipsoid: $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$, where (x,y,z) is the surface points and a,b,c are the radii on X,Y and Z directions respectively.

3. Elliptic parboiled: $\frac{x^2}{a^2}+\frac{y^2}{b^2}=z$

4. Hyperbolic parboiled : $\frac{x^2}{a^2}-\frac{y^2}{b^2}=z$

5. Elliptic cone : $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=0$

6. Hyperboloid of one sheet: $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$

7. Hyperboloid of two sheet: $\frac{x^2}{a^2}-\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$

- **Sphere**
  - In Cartesian coordinates, ***a spherical surface with radius r centered on the coordinate origin*** is defined ***as the set of points (x, y, z) that*** satisfy the equation

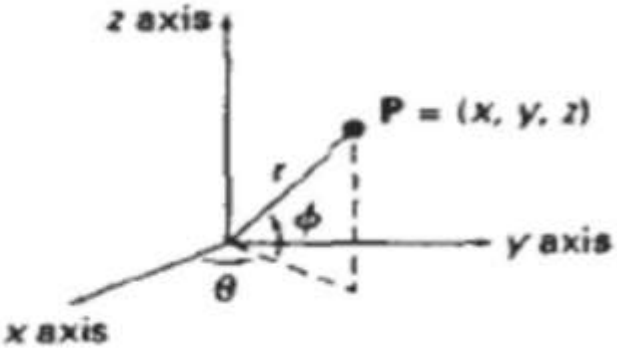
$$x^2 + y^2 + z^2 = r^2$$

  - ***The spherical surface in parametric form***

$$x = r\cos\phi\cos\theta, \qquad -\pi/2 \le \phi \le \pi/2$$

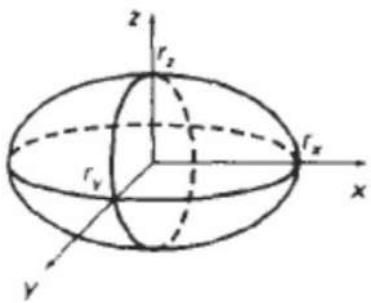
$$y = r\cos\phi\sin\theta, \qquad -\pi \le \theta \le \pi$$

$$z = r\sin\phi$$



- **Ellipsoidal**
  - An ellipsoidal surface can be described as an extension of a spherical surface, where the radii in three mutually perpendicular directions can have different values.
  - The Cartesian representation for points over the surface of an ellipsoid centered on the origin is $\left(\frac{x}{r_x}\right)^2 + \left(\frac{y}{r_y}\right)^2 + \left(\frac{z}{r_z}\right)^2 = 1$
  - And a parametric representation for the ellipsoid in terms of the latitude angle and the longitude angle

$$x = r_x \cos\phi \cos\theta, \qquad -\pi/2 \le \phi \le \pi/2$$

$$y = r_y \cos\phi \sin\theta, \qquad -\pi \le \theta \le \pi$$

$$z = r_z \sin\phi$$